Reg. No. | | | | | | | | | | | [23MSCH1H01]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2024

INORGANIC CHEMISTRY–I

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. Describe the physical properties of compounds with respect to intra- and intermolecular hydrogen bonding with examples.

b. Discuss the resonance structure of borazine.

c. Discuss the formation of covalent bonds in ClO_3^- anion using molecular orbital theory.

d. Define Arrhenius acid–base concept. Give examples.

e. **i.** What are nuclear forces? How do they differ from coulombic forces?

ii. Cobalt – 60, with a half-life of 5 years, is used in cancer radiation treatments. If a hospital purchases a supply of 30.0 g, how much will be left after 15 years?

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. **i.** List the postulates of molecular orbital theory.

ii. Explain the p-p and p-d combination of atomic orbitals.

b. Discuss the band theory of solids. How will you distinguish between intrinsic and extrinsic semiconductors?

c. **i.** Define lattice energy and describe the Born-Lande equation.

ii. Discuss the hybridization and structure of the following: SO_2, ClO_3F, XeF_4, IF_7, I_3^-.

SECTION – C

3. Answer the following question. **12**

a. **i.** Explain chain and pyro silicates with examples.

ii. Explain amphoteric nature of H_2O with chemical equation.

iii. Explain the analogy of the liquid drop model of nucleus. What are the similarities between nucleus and liquid drop?